

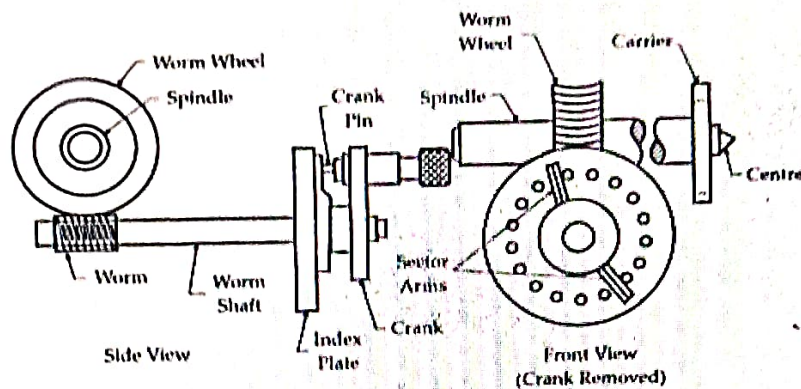
Machine Tool Technology Week-11

1. Explain Gear cutting by Indexing Methods
2. Prepare the Process plan as per the given drawing for Gear cutting
3. Estimate the Production cost per Piece as per the given drawing for Gear cutting

Indexing

Definition:

Indexing is the method of dividing the periphery of a piece of work into any number of equal parts. The attachment used for performing indexing is known indexing head. The indexing operation can be adapted for cutting gears, ratchet wheels, keyways, fluted drills, taps and reamers. The indexing head serves as an attachment for holding and indexing the work in doing the above tasks



There are three different types of Indexing heads namely,

1. Plain or simple dividing head IT is hand operated and used for simple indexing
2. Universal dividing head IT can be done by hand or gear drive depending on the operation
3. Optical dividing head. It is the most precision attachment and is therefore used for very precision indexing

Some German made Dividing heads, which are now quite commonly supplied with a large number of Milling machines manufactured in our country, are supplied with a set of 3 Index plates carrying hole circles are follows:

Plate No. 1

One side 13, 16, 18, 20, 23

Other side 15, 17, 19, 21, 24

Plate No. 2

One side 27, 28, 31, 37, 37, 41, 47

Other side 29, 33, 39, 43, 44, 49

Plat No. 3

One side 18 19, 20, 23, 29, 33, 39, 43, 49

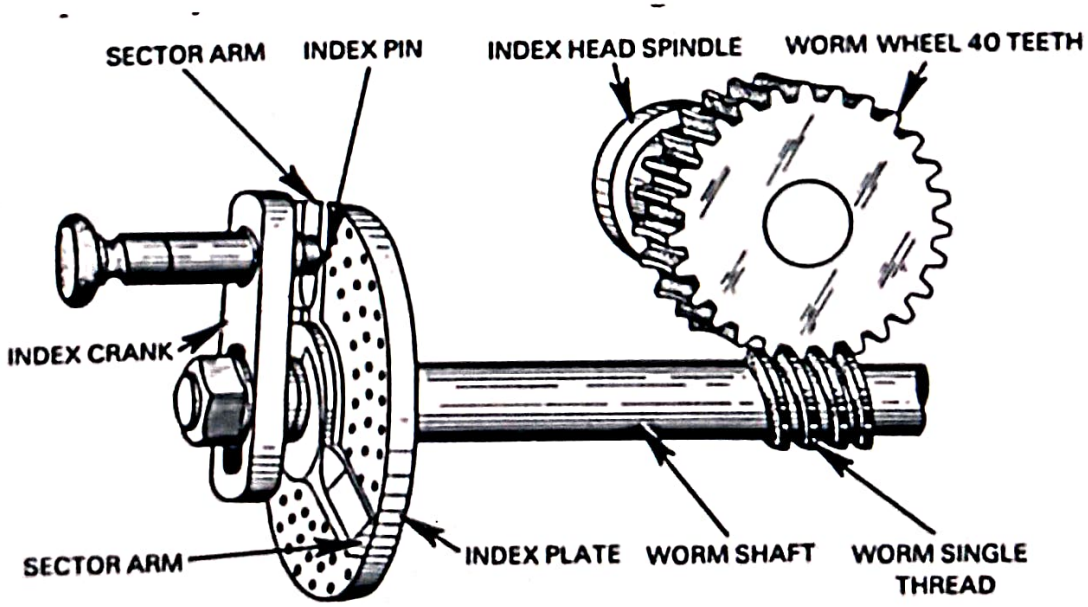
Other side 15, 17, 19, 21, 31, 37, 41, 47

Figure 1 The top plate is directly

ed by.

index plate

the crank as shown. When the



Flexing Mechanism

Example 11.3 It is required to divide the periphery of a job into 60 equal divisions. Find the Crank movement.

Solution. Required movement $= \frac{40}{60} = \frac{2}{3}$

Selecting 18 holes circle on plate No. 1, we get,

$$\frac{2}{3} = \frac{2}{3} \times \frac{6}{6} = \frac{12}{18}$$

i.e., 12 Holes on 18 Holes Circle.

Example 11.4 Required 35 divisions on a plate. Find the Indexing movement.

Solution. Required movement $= \frac{40}{35} = \frac{8}{7} = 1\frac{1}{7}$

Now, $\frac{1}{7} = \frac{1}{7} \times \frac{3}{3} = \frac{3}{21}$

i.e., 1 Full turn and 3 Holes on 21 Holes circle (Plate No. 2)

3. Compound Indexing:

In this type of indexing is employed in those cases when the number of divisions required on the workpiece cannot be obtained with the simple or plain index method.

This method involves two separate indexing movements that give the name compound indexing method.

This is performed in two stages:

1. The first movement is obtained by turning the work a definite amount in one direction in same way as in simple indexing.
2. The second index movement is obtained by turning the indexing plate along the crank.

Example 11.6 Compound Indexing for 51 divisions.

Solution. Required movement = $\frac{40}{51}$

Let us try circles of 17 and 18 holes.

The first expression = $\frac{3 \times 17 \times 1}{10 \times 4 \times 17 \times 3 \times 6} = \frac{1}{240}$

We get unity in numerator, i.e., the circles selected are correct.

Now the Required movements is given by :

$$\begin{aligned} \frac{240}{17} - \frac{240}{18} \quad \text{or} \quad \frac{240}{18} - \frac{240}{17} \\ = 14 \frac{2}{17} - 3 \frac{6}{18} \quad \text{or} \quad 13 \frac{6}{18} - 14 \frac{2}{17} \end{aligned}$$

By taking out 14 as common, the above expression will be reduced as :

$$= \frac{2}{17} + \frac{12}{18} \quad \text{or} \quad -\frac{12}{18} - \frac{2}{17}$$

Similar signs show that both the movements will be in the same direction. By adopting the first result we get the Required Movement as :

Crank 2 holes on 17 holes circle and Crank and plate 12 holes on 18 holes circle ; both in the same direction.

By applying the Second check :

$$\text{Crank movements} = \frac{2}{17} + \frac{12}{18} \quad \text{or} \quad \frac{40}{51} = \frac{40}{N}$$

Hence, the obtained result is correct.

4. Differential Indexing:

This is similar to compound indexing where the required division of the periphery of job is obtained by the combination of two movements.

- 1) The movement of Indexing crank through the required number of spaces in one of whole circle of index plate as in case of simple indexing.
- 2) The simultaneous movement of index plate along with cranks either in same direction or reverses direction.